تجمع جازان الصحي
Jazan Health Cluster

King Fahad Central Hospital

Aligned with AHA/ASA Guidelines for Early Management of Acute Ischemic Stroke and pivotal endovascular trial eligibility criteria (MR CLEAN/ESCAPE/SWIFT PRIME/EXTEND-IA/REVASCAT; DAWN; DEFUSE 3)

PART 1 — Patient Selection Protocol

1.1 General Eligibility (applies to all candidates)

☐ Age ≥ 18 years

☐ Acute ischemic stroke with disabling neurologic deficit

☐ Large vessel occlusion (LVO) confirmed on CT angiography or MR angiography — intracranial ICA, MCA-M1 (and selected M2), or basilar artery

☐ Pre-stroke functional status independent or near-independent (modified Rankin Scale 0–1 for standard trials; individualize for higher baseline mRS)

☐ Groin/arterial puncture can be achieved within the eligible time window from last known well

☐ No contraindication to endovascular intervention (see Section 1.5)

Concurrent IV thrombolysis should be given to eligible patients without delaying thrombectomy — do not withhold or delay EVT to await IV alteplase/tenecteplase response.

1.2 Early Window: 0–6 hours from Last Known Well

Criteria drawn from the pivotal 2015 trials (MR CLEAN, ESCAPE, SWIFT PRIME, EXTEND-IA, REVASCAT) as codified in AHA/ASA Class I recommendations.

Criterion	Threshold
Pre-stroke mRS	0–1
Occlusion site	Causative occlusion of intracranial ICA or MCA segment 1 (M1)
NIHSS	≥ 6
ASPECTS (non-contrast CT)	≥ 6
Time to treatment	Groin puncture achievable within 6h of LKW

1.3 Extended Window: 6–16 hours (DEFUSE 3–based criteria)

Criterion	Threshold
Age	18–90
Occlusion site	Intracranial ICA or MCA-M1
NIHSS	≥ 6
Pre-stroke mRS	0–2
Ischemic core volume (perfusion imaging)	< 70 mL
Mismatch ratio (penumbra:core)	≥ 1.8
Mismatch volume	≥ 15 mL

1.4 Extended Window: 6–24 hours (DAWN-based clinical-imaging mismatch criteria)

Group	Age	NIHSS	Core infarct volume (perfusion/DWI)
A	≥ 80	≥ 10	< 21 mL
B	< 80	≥ 10	< 31 mL
C	< 80	≥ 20	31–51 mL

Occlusion: intracranial ICA or MCA-M1. Patients meeting either DAWN or DEFUSE 3 criteria are reasonable candidates in the 6–24h window per AHA/ASA Class I/IIa recommendations — confirm current guideline class/level at time of use.

1.5 Additional / Evolving Considerations — Confirm Against Current Guideline

☐ M2 (and selected distal medium-vessel) occlusions: thrombectomy may be reasonable in select patients with significant deficit — individualize (evidence lower certainty than M1/ICA)
☐ Basilar artery occlusion: EVT may be considered up to 24h in selected patients per evolving evidence (e.g., ATTENTION, BAOCHE) — confirm this has been incorporated into current institutional/AHA guideline version before relying on it
☐ Large core infarct (ASPECTS 3–5 or larger volumes): thrombectomy may be reasonable in select patients per newer large-core trials — individualize with neurointerventional/stroke attending, confirm current guideline stance
☐ Mild deficit (NIHSS <6) with LVO: not established first-line; individualize
☐ Tandem occlusion (cervical ICA + intracranial LVO): may require concurrent extracranial stenting — per interventionalist judgment

1.6 Contraindications / Exclusions to Review

☐ Pre-existing severe disability (baseline mRS precluding meaningful functional recovery) — individualize
☐ Evidence of large completed infarct incompatible with intervention on imaging (per interventionalist/neuroradiology review)
☐ Active serious hemorrhage or known bleeding diathesis precluding femoral/radial access
☐ Inability to obtain vascular access
☐ Life expectancy/goals of care inconsistent with aggressive intervention (confirm with patient/surrogate and care team)
☐ Contrast allergy — pre-medicate per institutional protocol if intervention still indicated

1.7 Pre-Procedure Checklist

☐ CTA/MRA confirms target occlusion; non-contrast CT ASPECTS or perfusion imaging obtained per window
☐ NIHSS documented
☐ Labs: platelet count, INR/aPTT, glucose, renal function (Cr/eGFR) for contrast risk stratification
☐ Weight-based IV thrombolysis given if eligible, without delaying transfer to angiography suite
☐ Informed consent obtained from patient/surrogate (or emergent consent per institutional policy)
☐ Anesthesia/sedation plan (conscious sedation vs. general anesthesia) per interventionalist and anesthesia team
☐ Stroke/Neurointervention team and angiography suite notified — target door-to-puncture time per institutional metric

PART 2 — Post-Thrombectomy Care Order Set

Patient Name: ______________________ **DOB:** _________ MRN: ____________	**Last Known Well (LKW):** ___________ **Groin/Access Puncture Time:** ______
Interventionalist: _______________	**Access Site:** Femoral / Radial

Reperfusion grade (mTICI): 0 1 2a 2b 2c 3 (circle)

2.1 Disposition & Monitoring

☐ Admit to Neuro ICU / dedicated stroke unit per institutional protocol and clinical status
☐ Neuro checks + vital signs: q15min x 1–2h, then q30min x 6h, then q1h x 16h (adjust per unit protocol)
☐ Continuous cardiac (telemetry) monitoring
☐ Continuous pulse oximetry; supplemental O2 to keep SpO2 > 94%
☐ Notify physician for: new/worsening neurologic deficit, NIHSS increase ≥2 points, decreased LOC, severe headache, SBP/DBP outside target range, access-site bleeding or limb ischemia signs

2.2 Access-Site Care

☐ Femoral access: check groin site and distal pulses q15min x1h, q30min x2h, then q1h x4h (or per closure device instructions)
☐ Bed rest / limb immobilization per closure method: manual compression — flat bed rest per protocol (commonly 4–6h); closure device — per device instructions (commonly shorter)
☐ Radial access: check radial pulse and hand perfusion per protocol; hemostasis band deflation per radial protocol
☐ Assess for hematoma, retroperitoneal bleed (flank/back pain, hypotension, falling hematocrit), pseudoaneurysm, distal limb ischemia
☐ Foley catheter (if placed) removed as soon as clinically appropriate to reduce infection risk

2.3 Blood Pressure Management

☐ Maintain SBP < 180 mmHg and DBP < 105 mmHg for first 24 hours post-procedure (standard target)
☐ Consider individualized lower target (e.g., SBP <160 mmHg) if successful reperfusion (mTICI 2b–3) — per treating physician, weigh against reperfusion/hemorrhage risk
☐ Avoid hypotension — maintain adequate cerebral perfusion, especially if incomplete reperfusion
☐ Antihypertensive agent/route per protocol: __________

2.4 Neuroimaging

☐ Non-contrast head CT immediately post-procedure per institutional protocol to assess for hemorrhage
☐ Repeat non-contrast head CT (or MRI) at approximately 24 hours, or sooner for any neurologic decline
☐ Monitor for hemorrhagic transformation, malignant edema — consider neurosurgical evaluation for decompressive hemicraniectomy if large-territory infarct with edema, especially within 24–48h

2.5 Antithrombotic / Anticoagulation Timing

☐ Hold antiplatelet and anticoagulant agents until post-procedure hemorrhage has been excluded on repeat imaging
☐ If IV thrombolysis was given: hold antithrombotics until 24h post-tPA and hemorrhage excluded, per standard post-tPA protocol
☐ Once hemorrhage excluded: initiate antiplatelet or anticoagulation per stroke etiology — physician to specify agent, dose, and timing
☐ If stent placed (e.g., for tandem occlusion or rescue stenting): dual antiplatelet therapy per interventionalist protocol

2.6 Labs, Renal & Contrast Considerations

☐ Basic metabolic panel / renal function post-procedure — monitor for contrast-induced nephropathy
☐ IV hydration per protocol if renal function impaired or large contrast load given
☐ CBC, coagulation studies as clinically indicated

2.7 General Supportive Care (as for ischemic stroke)

☐ NPO until formal dysphagia screen completed; diet advanced per screen result
☐ HOB elevated ≥30° once stable
☐ Fingerstick glucose monitoring; treat to target 140–180 mg/dL
☐ Antipyretic for temperature >38.0°C — maintain normothermia
☐ VTE prophylaxis: sequential compression devices; pharmacologic prophylaxis once hemorrhage excluded and per bleeding risk
☐ High-intensity statin therapy unless contraindicated
☐ Fall and aspiration precautions

2.8 Consults & Multidisciplinary Care

☐ Neurology / Stroke service
☐ Physical Therapy, Occupational Therapy
☐ Speech-Language Pathology (swallow and communication evaluation)
☐ Case Management / Social Work — discharge planning, rehab placement
☐ Neurosurgery, if hemorrhagic conversion or malignant edema
☐ Vascular surgery, if access-site complication requiring intervention

2.9 Quality & Registry Metrics to Document

☐ Last known well time, arrival time, groin/access puncture time, recanalization time
☐ Door-to-puncture and puncture-to-recanalization intervals
☐ Final mTICI reperfusion grade
☐ NIHSS at 24h and at discharge
☐ Symptomatic intracranial hemorrhage (yes/no)
☐ Modified Rankin Scale at 90-day follow-up (registry/Get With The Guidelines–Stroke reporting)

Ordering Physician / Interventionalist Authorization

Physician Signature: ______________________________ Date/Time: ____________________

Print Name: ______________________________ Pager/Contact: __________________

References: 2019 AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; DAWN trial (Nogueira et al., NEJM 2018); DEFUSE 3 trial (Albers et al., NEJM 2018); MR CLEAN, ESCAPE, SWIFT PRIME, EXTEND-IA, REVASCAT (2015); AHA/ASA Get With The Guidelines–Stroke registry measures; The Joint Commission Comprehensive/Thrombectomy-Capable Stroke Center certification requirements. This template should be re-verified against the current guideline version at the time of institutional adoption, as endovascular eligibility criteria (basilar occlusion, large core, distal occlusions) continue to evolve.